```
DROP DATABASE IF EXISTS adaptive_learning;

CREATE DATABASE adaptive_learning
CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

USE adaptive_learning;

CREATE TABLE users (
    id          INT AUTO_INCREMENT PRIMARY KEY,
    name        VARCHAR(120)       NOT NULL,
    email       VARCHAR(255) UNIQUE NOT NULL,
    password_hash VARCHAR(255)       NOT NULL,
    role        ENUM('student','instructor','admin') DEFAULT 'student',
    avatar_url    VARCHAR(500),
    created_at    DATETIME DEFAULT CURRENT_TIMESTAMP,
    updated_at    DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);

CREATE TABLE courses (
    id          INT AUTO_INCREMENT PRIMARY KEY,
    title       VARCHAR(255)  NOT NULL,
    description   TEXT,
    thumbnail_url VARCHAR(500),
    difficulty    ENUM('beginner','intermediate','advanced') DEFAULT 'beginner',
    instructor_id INT NOT NULL,
    is_published  BOOLEAN DEFAULT FALSE,
    created_at    DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (instructor_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE modules (
```

```
    id       INT AUTO_INCREMENT PRIMARY KEY,
    course_id INT NOT NULL,
    title     VARCHAR(255) NOT NULL,
    order_num INT DEFAULT 0,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE CASCADE
);

CREATE TABLE lessons (
    id       INT AUTO_INCREMENT PRIMARY KEY,
    module_id INT NOT NULL,
    title     VARCHAR(255) NOT NULL,
    content   LONGTEXT,
    video_url VARCHAR(500),
    order_num INT DEFAULT 0,
    xp_reward INT DEFAULT 10,
    FOREIGN KEY (module_id) REFERENCES modules(id) ON DELETE CASCADE
);

CREATE TABLE enrollments (
    id         INT AUTO_INCREMENT PRIMARY KEY,
    user_id     INT NOT NULL,
    course_id   INT NOT NULL,
    enrolled_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    UNIQUE KEY uq_enroll (user_id, course_id),
    FOREIGN KEY (user_id)   REFERENCES users(id)   ON DELETE CASCADE,
    FOREIGN KEY (course_id) REFERENCES courses(id) ON DELETE CASCADE
);

CREATE TABLE lesson_progress (
    id          INT AUTO_INCREMENT PRIMARY KEY,
    user_id      INT NOT NULL,
```

```
    lesson_id    INT NOT NULL,
    completed    BOOLEAN DEFAULT FALSE,
    time_spent_s INT DEFAULT 0,
    completed_at DATETIME,
    UNIQUE KEY uq_lp (user_id, lesson_id),
    FOREIGN KEY (user_id)   REFERENCES users(id)   ON DELETE CASCADE,
    FOREIGN KEY (lesson_id) REFERENCES lessons(id) ON DELETE CASCADE
);

CREATE TABLE questions (
    id          INT AUTO_INCREMENT PRIMARY KEY,
    lesson_id   INT NOT NULL,
    question    TEXT NOT NULL,
    type        ENUM('mcq','true_false','fill') DEFAULT 'mcq',
    options     JSON,
    correct     VARCHAR(255) NOT NULL,
    explanation TEXT,
    difficulty  TINYINT DEFAULT 1,
    FOREIGN KEY (lesson_id) REFERENCES lessons(id) ON DELETE CASCADE
);

CREATE TABLE quiz_attempts (
    id          INT AUTO_INCREMENT PRIMARY KEY,
    user_id     INT NOT NULL,
    lesson_id   INT NOT NULL,
    score       FLOAT DEFAULT 0,
    total_q     INT DEFAULT 0,
    correct_q   INT DEFAULT 0,
    answers     JSON,
    attempted_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id)   REFERENCES users(id)   ON DELETE CASCADE,
```

```
    FOREIGN KEY (lesson_id) REFERENCES lessons(id) ON DELETE CASCADE
);

CREATE TABLE learner_profiles (
    id            INT AUTO_INCREMENT PRIMARY KEY,
    user_id       INT UNIQUE NOT NULL,
    skill_level   FLOAT DEFAULT 0.5,
    learning_speed FLOAT DEFAULT 1.0,
    weak_topics   JSON,
    strong_topics JSON,
    preferred_type ENUM('visual','reading','practice') DEFAULT 'reading',
    total_xp      INT DEFAULT 0,
    streak_days   INT DEFAULT 0,
    last_active   DATETIME,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE badges (
    id            INT AUTO_INCREMENT PRIMARY KEY,
    name          VARCHAR(120) NOT NULL,
    description   TEXT,
    icon          VARCHAR(10),
    condition_text VARCHAR(255)
);

CREATE TABLE user_badges (
    id       INT AUTO_INCREMENT PRIMARY KEY,
    user_id  INT NOT NULL,
    badge_id INT NOT NULL,
    earned_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    UNIQUE KEY uq_ub (user_id, badge_id),
```

```
    FOREIGN KEY (user_id)  REFERENCES users(id)   ON DELETE CASCADE,
    FOREIGN KEY (badge_id) REFERENCES badges(id)  ON DELETE CASCADE
);
INSERT INTO badges (name, description, icon, condition_text) VALUES
('First Step',    'Completed your first lesson',        '🎯', 'lessons_completed=1'),
('Quiz Master',   'Scored 100% on a quiz',              '🏆', 'perfect_quiz=1'),
('Speed Learner', 'Completed 5 lessons in one day',     '⚡', 'daily_lessons=5'),
('Consistent',    '7-day learning streak',              '🔥', 'streak=7'),
('Scholar',       'Earned 500 XP',                      '📚', 'xp=500');

INSERT INTO users (name, email, password_hash, role) VALUES
('Nakul',       'nakularora697@gmail.com',
'615f568962ccf748113f4d3b714de8454ca13d07b4df62f331a3c1311fa860ba', 'admin'),
('Admin',       'admin@learn.io',
'e138e47ad2fb061de54cc449c2c0dfd40622e1c395e49c5a1f3d82c07532c881', 'admin'),
('Rohan Sharma', 'rohan1@gmail.com',
'e138e47ad2fb061de54cc449c2c0dfd40622e1c395e49c5a1f3d82c07532c881', 'student'),
('Ankit Verma',  'ankit1@gmail.com',
'e138e47ad2fb061de54cc449c2c0dfd40622e1c395e49c5a1f3d82c07532c881', 'student'),
('Priya Singh',  'priya1@gmail.com',
'e138e47ad2fb061de54cc449c2c0dfd40622e1c395e49c5a1f3d82c07532c881', 'student'),
('Dr. Mehta',    'mehta1@gmail.com',
'e138e47ad2fb061de54cc449c2c0dfd40622e1c395e49c5a1f3d82c07532c881', 'instructor');

INSERT INTO learner_profiles
    (user_id, skill_level, learning_speed, weak_topics, strong_topics, preferred_type, total_xp,
streak_days)
SELECT id, 0.5, 1.0, '[]', '[]', 'reading', 0, 0 FROM users;

INSERT INTO courses (title, description, difficulty, instructor_id, is_published) VALUES
('Python Basics', 'Learn Python from scratch with hands-on examples', 'beginner',     6, TRUE),
('Web Dev',       'Master HTML, CSS and JavaScript',                  'beginner',     6, TRUE),
('Data Science',  'Introduction to Data Science and ML',              'intermediate', 6, TRUE);
```

INSERT INTO modules (course_id, title, order_num) VALUES

(1, 'Introduction to Python', 1),

(1, 'Control Flow', 2),

(2, 'HTML Fundamentals', 1),

(2, 'CSS Styling', 2),

(3, 'Data Analysis', 1);

LESSONS

INSERT INTO lessons (module_id, title, content, order_num, xp_reward) VALUES

(1, 'What is Python',

'## What is Python?\n\nPython is a high-level, interpreted programming language.\n\n**Key Features:**\n- Easy to read\n- Large standard library\n- Cross-platform\n\n```python\nprint("Hello World")\n```',

1, 20),

(1, 'Variables',

'## Variables in Python\n\nA variable stores a value.\n\n```python\nname = "Nakul"\nage = 20\nprint(name, age)\n```\n\n**Types:** int, float, str, bool',

2, 20),

(2, 'For Loop',

'## For Loops\n\nUsed to iterate over a sequence.\n\n```python\nfor i in range(5):\n print(i)\n```\n\nOutput: 0 1 2 3 4',

1, 25),

(3, 'HTML Tags',

'## HTML Tags\n\nHTML uses tags to structure content.\n\n```html\n<h1>Heading</h1>\n<p>Paragraph</p>\n<a href="#">Link</a>\n```',

1, 20),

(4, 'CSS Colors',

'## CSS Colors\n\nCSS controls the visual style.\n\n```css\nbody { background-color: #0b0d17; color: white; }\nh1 { color: #6c63ff; }\n```',

1, 20),

(5, 'Pandas Basics',

'## Pandas\n\nPandas is used for data manipulation.\n\n```python\nimport pandas as pd\ndf = pd.read_csv("data.csv")\nprint(df.head())\n```',

1, 30);

QUESTIONS

INSERT INTO questions (lesson_id, question, type, options, correct, explanation, difficulty) VALUES

(1, 'What type of language is Python?',

 'mcq', '["Compiled","Interpreted","Assembly","Machine"]',

 'Interpreted', 'Python is executed line by line by an interpreter.', 1),

(1, 'Python is case-sensitive',

 'true_false', NULL,

 'True', 'Python treats Name and name as different variables.', 1),

(2, 'Which symbol assigns a value in Python?',

 'mcq', '["==","=>","=","::"]',

 '=', 'The = operator assigns a value to a variable.', 1),

(2, 'What will type(42) return?',

 'mcq', '["<class str>","<class float>","<class int>","<class num>"]',

 '<class int>', '42 is an integer literal in Python.', 2),

(3, 'What does range(3) produce?',

 'mcq', '["1,2,3","0,1,2","0,1,2,3","1,2"]',

 '0,1,2', 'range(3) generates 0 up to but not including 3.', 1),

(4, 'Which tag is the largest heading in HTML?',

```
 'mcq', '["<h6>","<heading>","<h1>","<head>"]',
 '<h1>', 'h1 is the largest heading tag in HTML.', 1),

(4, 'HTML stands for?',
 'fill', NULL,
 'HyperText Markup Language', 'HTML = HyperText Markup Language.', 1),

(5, 'Which CSS property changes text color?',
 'mcq', '["font-color","text-color","color","foreground"]',
 'color', 'The color property sets the text color in CSS.', 1),

(6, 'Which Python library is used for data manipulation?',
 'mcq', '["numpy","pandas","matplotlib","scipy"]',
 'pandas', 'Pandas provides DataFrame and Series for data analysis.', 1);

INSERT INTO enrollments (user_id, course_id) VALUES
(3, 1), (4, 1), (5, 2), (3, 2), (4, 3);

-- ── VERIFY ──────────────────────────────────────────────────────────
SELECT id, name, email, role FROM users;
```

id	name	email	role
1	Nakul	nakularora697@gmail.com	admin
2	Admin	admin@learn.io	admin
3	Rohan Sharma	rohan1@gmail.com	student
4	Ankit Verma	ankit1@gmail.com	student
5	Priya Singh	priya1@gmail.com	student
6	Dr. Mehta	mehta1@gmail.com	instructor

SELECT id, title, is_published FROM courses;

id	title	is_published
1	Python Basics	1
2	Web Dev	1
3	Data Science	1
NULL	NULL	NULL

SHOW TABLES;

Tables_in_adaptive_learning
badges
courses
enrollments
learner_profiles
lesson_progress
lessons
modules
questions
quiz_attempts
user_badges
users